ISEP

**Segurança de Sistemas e Informação (SEGSI)**
PL02
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## Exercise 1: TACACS+

TACACS is a protocol often used to authenticate administrator to routers and switches. It's based on a secret key to authenticate and is considered no longer secure. From time to time, we see this deployed in old corporate networks.

1. Install TACACS+ package in a Linux Virtual Machine;
2. Configure the package with a secret key and serve it;
3. Create users on your machine: Bob, Alice, John, Madeleine, and Jake. Bob and Alice are standard users, Madeleine is an accountant, and Jake is the the "IT Guy" (Administrator). Create the necessary groups;
4. Capture the traffic and try to extract and crack your own key (suggestion: use the hcxcaptool[1])
5. What are your security considerations for this protocol?

## Exercise 2: Diameter

RADIUS is one of the most known protocols to provide AAA services. It is often seen in Wireless Infrastructure where 802.1X enabled networks allow users to connect to their corporate credentials to the network providing easy account management of the Wireless infrastructure itself. Diameter is a protocol that brings new security features to the AAA scheme. It shares some similarities with RADIUS but offers several improvements. For instance, Radius Authentication is client-server oriented, however, Diameter can invert this logic if it needs to.

1. Install the FreeDiameter package on your server;
2. Configure it to server authentication to clients using a secure connection;
3. Diameter is often tied with new generation networks such as 5G. What extra capabilities do you see fit in alignment with the concept of "Secure Networks"?

[1] https://github.com/warecrer/Hcxpcaptool